

Changes in nutrient intake and dietary quality among participants with type 2 diabetes following a low-fat vegan diet or a conventional diabetes di...

BACKGROUND: Although vegan diets improve diabetes management, little is known about the nutrient profiles or diet quality of individuals with type 2 diabetes who adopt a vegan diet. **OBJECTIVE:** To assess the changes in nutrient intake and dietary quality among participants following a low-fat vegan diet or the 2003 American Diabetes Association dietary recommendations. **DESIGN:** A 22-week randomized, controlled clinical trial examining changes in nutrient intake and diet quality. **SUBJECTS/SETTING:** Participants with type 2 diabetes (n=99) in a free-living setting. **RESEARCH DESIGN AND METHODS:** Participants were randomly assigned to a low-fat vegan diet or a 2003 American Diabetes Association recommended diet. **MAIN OUTCOME MEASURES:** Nutrient intake and Alternate Healthy Eating Index (AHEI) scores were collected at baseline and 22 weeks. **STATISTICAL ANALYSES PERFORMED:** Between-group t tests were calculated for changes between groups and paired comparison t tests were calculated for changes within-group. Pearson's correlation assessed relationship of AHEI score to hemoglobin A1c and body weight changes. **RESULTS:** Both groups reported significant decreases in energy, protein, fat, cholesterol, vitamin D, selenium, and sodium intakes. The vegan group also significantly reduced reported intakes of vitamin B-12 and calcium, and significantly increased carbohydrate, fiber, total vitamin A activity, beta carotene, vitamins K and C, folate, magnesium, and potassium. The American Diabetes Association recommended diet group also reported significant decreases in carbohydrate and iron, but reported no significant increases. The vegan group significantly improved its AHEI score ($P<0.0001$), while the American Diabetes Association recommended diet group did not ($P=0.7218$). The difference in AHEI score at 22 weeks between groups was significant ($P<0.0001$). With both groups combined, AHEI score was negatively correlated with both changes in hemoglobin A1c value ($r=-0.24$, $P=0.016$) and weight ($r=-0.27$, $P=0.007$). **CONCLUSIONS:** Vegan diets increase intakes of carbohydrate, fiber, and several micronutrients, in contrast with the American Diabetes Association recommended diet. The vegan group improved its AHEI score whereas the American Diabetes Association recommended diet group's AHEI score remained unchanged.